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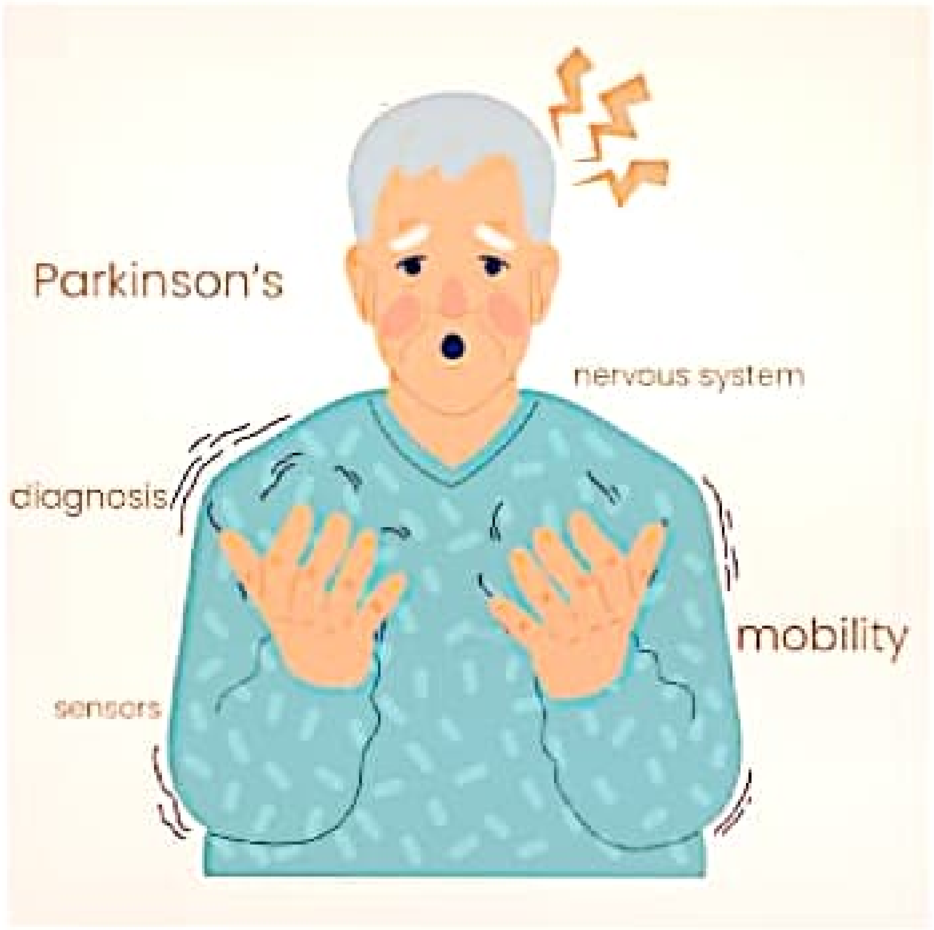
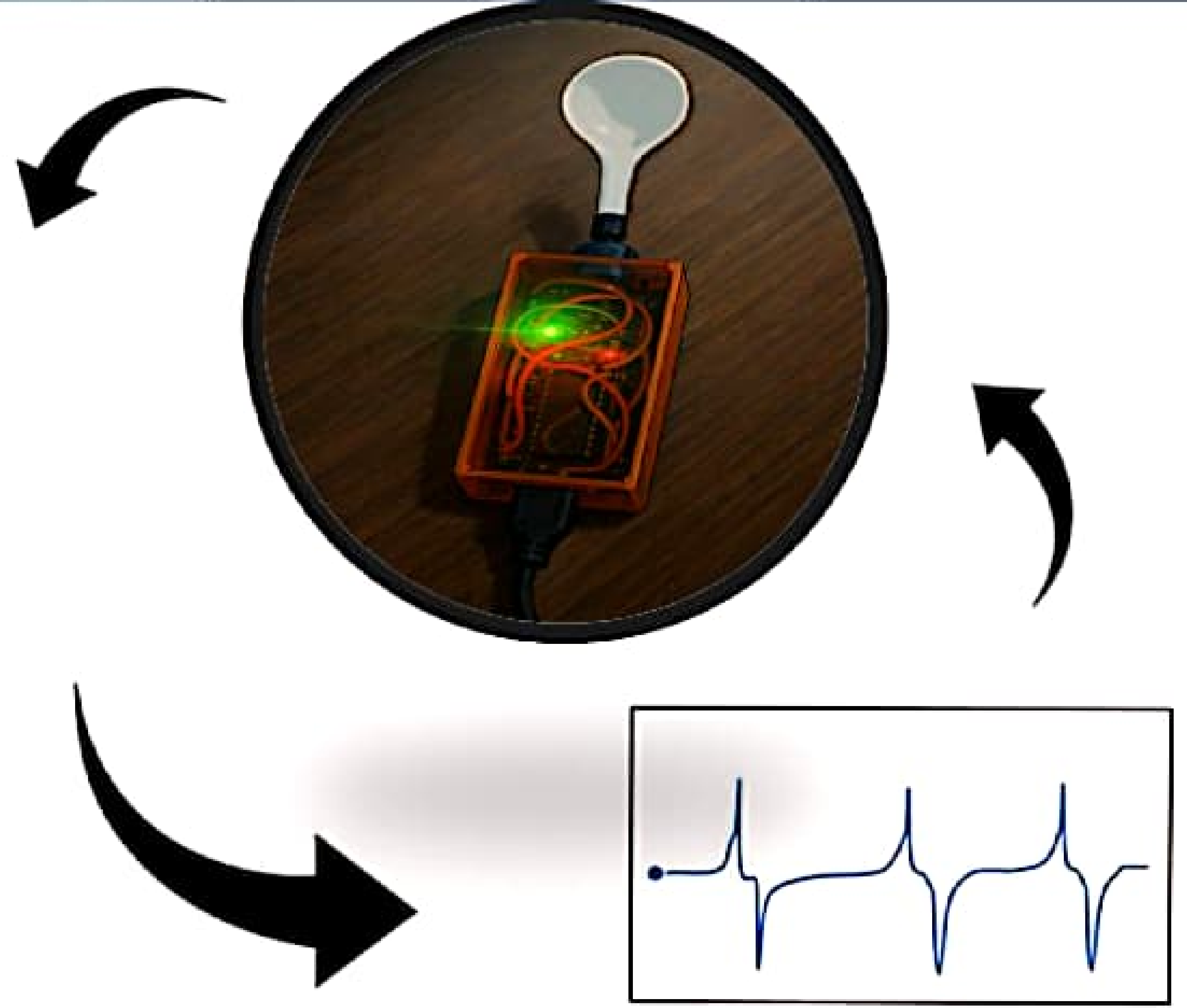




Design and Development of Parkinsons's Patient Spoon

Motivation

- Parkinson's disease causes involuntary hand tremors that make simple tasks like eating extremely difficult for patients.

- Loss of hand stability reduces independence and affects the emotional and physical well-being of individuals.

- There is a strong need for an **affordable, user-friendly, and reliable** solution that can help Parkinson's patients regain dignity and independence during eating.

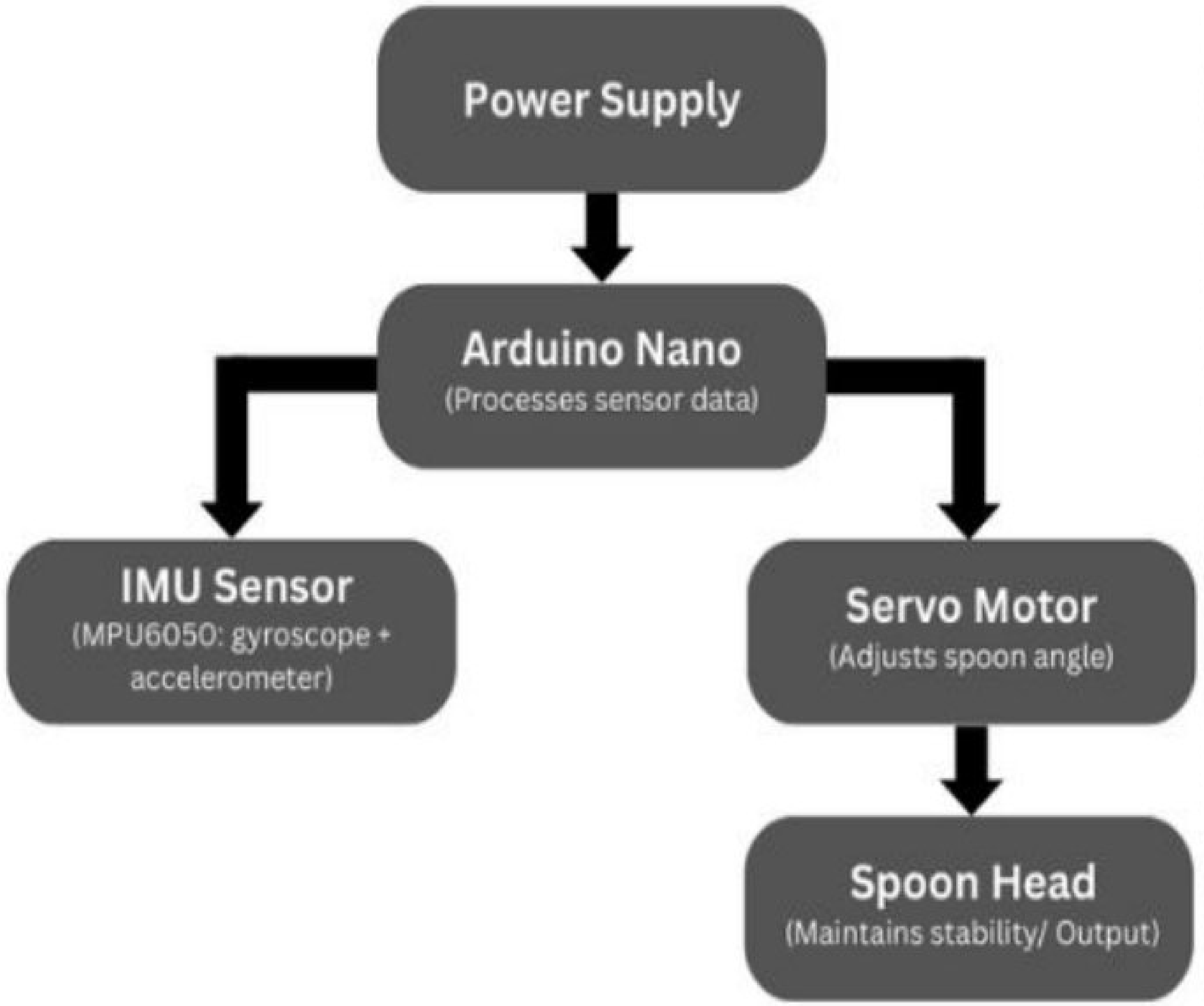
- Our motivation is to design a **low-cost self-stabilizing spoon** that compensates for tremors without compromising usability, accessibility, or performance.

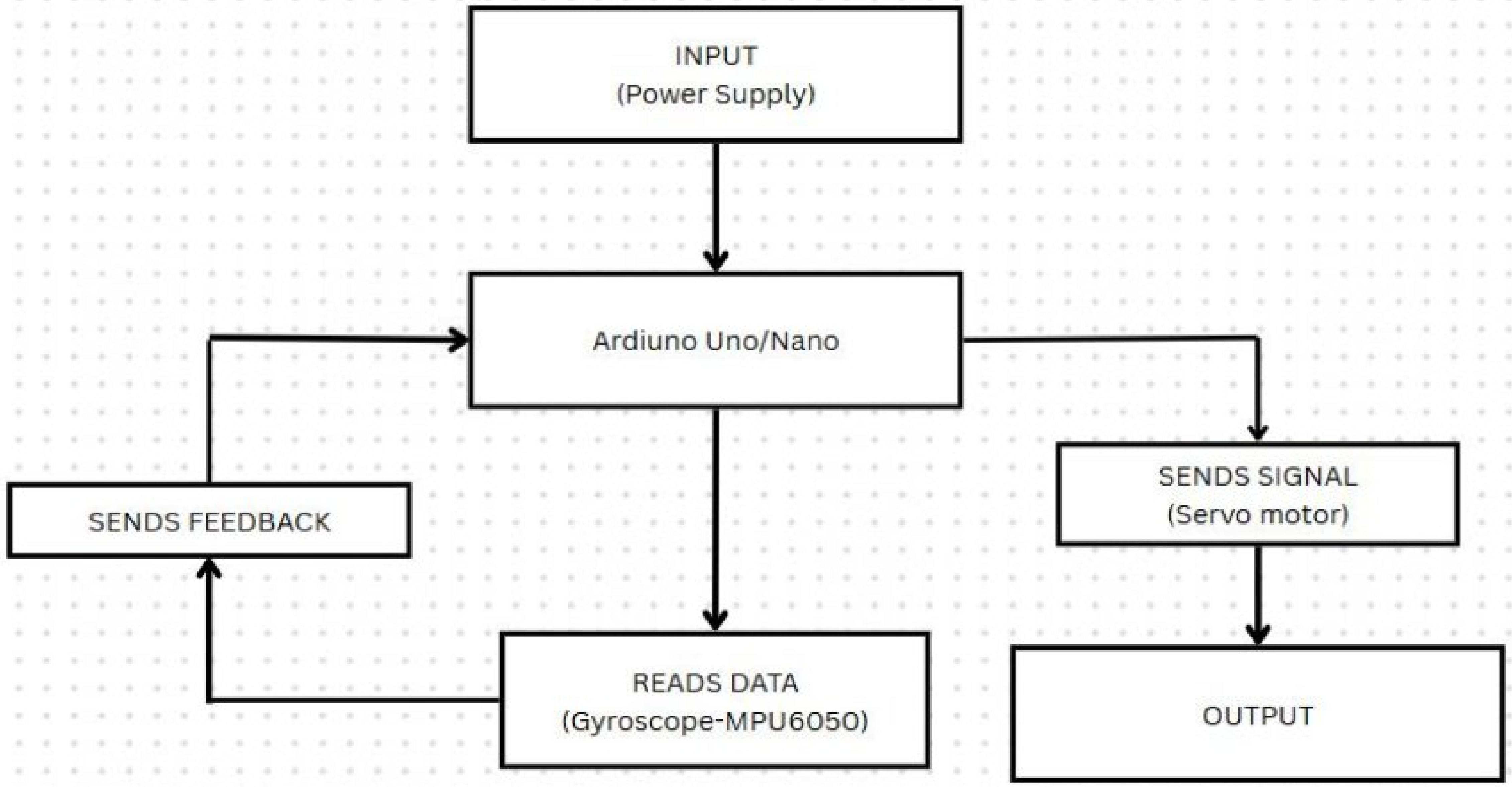
Proposed Solution

Development of spoon for Parkinson's Patients

A self-stabilizing spoon equipped with motion sensors and servo motors, managed by a microcontroller, is proposed. The system compensates for the user's hand tremors by maintaining the spoon's bowl in a horizontal and stable position, regardless of involuntary hand movements. This device is designed to offer a comparable performance to market solutions but at a fraction of the cost.

Approach





Block diagram of the system

Components Used

Nano Arduino

MPU 6050 sensor

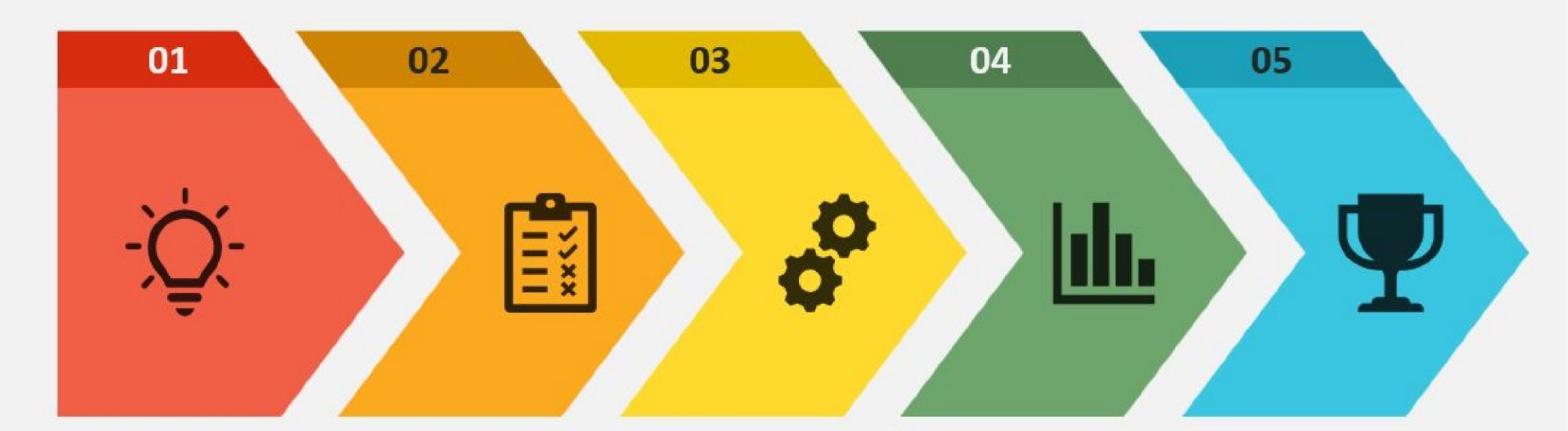
Servo Motor

ARDUINO
IDE

Kapton sheet

Connecting Wire

WORKFLOW



A **gyroscope sensor** (gyro) detects hand tremors and unwanted movements in real-time.

The **Arduino Nano** reads gyro data and processes it to determine the direction and amount of tremor.

Based on the processed data, the **servo motor** adjusts the spoon's position to counteract the tremor, stabilizing it.

External Power source supplies power to the Arduino .

We developed a self balancing spoon with the logic that if **we tilt our handle by angle α then spoon tilts -α**.

Code Overview

1. Complementary Filter (Sensor Fusion)

Purpose: To accurately calculate the tilt (pitch) angle using both accelerometer and gyroscope data.

Concept:

- Accelerometer gives stable but noisy angle data.
- Gyroscope gives smooth data but drifts over time.
- A complementary filter combines both:
  angle = **α × (angle + gyroRate × dt) + (1 - α) × accelAngle**

2.PD Controller (Proportional-Derivative Control)

Purpose: To calculate how much the servo should move to correct the tilt.

Concept:

- **Proportional (P)**: Corrects based on current tilt angle.
- **Derivative (D)**: Adds damping based on how fast the angle is changing.
- Formula: control = **Kp × error + Kd × (error change / time)**
- Helps achieve fast and stable correction without overshooting.

3.Servo Control (Actuation)

Purpose: To physically move the servo based on the PD controller's output.

Concept:

- Servo is centered at 90°.
- Control output is subtracted from 90 to move in the opposite direction of tilt.
- Ensures the spoon tilts back to balance.
- Formula: servoPosition = **constrain(90 - control, 0, 180)**

Challenges Encountered During System Development

Directional Inconsistency in Servo Response
Servo moved correctly along one axis but incorrectly along another due to improper mapping between sensor axes and motor control logic.

Calibration Errors:
Misalignment during MPU6050 initialization resulted in incorrect neutral positions, reducing stabilization performance. That is why we incorporated a Complementary Filter

3D printing
Initially assembly was done on bread board which occupied a lot of space.Later we soldered everything and made a compact Design

Burnout of Arduino Nano
The Arduino Burnt out during testing so for demonstration we are using Arduino.

Results

Future Work

- Eliminate wired connections between the stabilizing mechanism and power source.
- Implement wireless charging (like magnetic docks or Qi wireless) for easier recharging.
- Use Bluetooth/Wi-Fi connectivity to monitor device performance or customize settings via a mobile app.
- Design a more **comfortable, non-slip grip** using soft-touch, medical-grade materials like silicone.
- Create easily swappable attachments like a **fork, knife, or soup spoon** using magnetic or twist-lock systems.

Thank you

For your attention!

Questions and suggestions are welcome ☺️